

Practical Lab Guide

Security, Automation, & Low-Voltage Systems Training

Comprehensive Practical Syllabus for 28 Hands-On Sessions

Module 1: Electrical & Networking Basics (Sessions 1 - 13)

Module 2: Professional CCTV Systems (Sessions 14 - 28)

Syllabus Scope: Step-by-Step Lab Plans, Consumable Materials, & Essential Teaching Tools.

Master Tools & Instrumentation Inventory

Consolidated shopping list of all teaching tools required for the instructor and student workstations across the 28 sessions.

Category	Item Description	Primary Sessions Used
Hand Tools	Insulated Screwdrivers (Set of Phillips & Flathead 1000V rated), Precision flathead terminal screwdrivers, heavy-duty wire strippers, utility knife.	Sessions 1, 2, 4, 5, 6, 7, 13, 17, 21, 22, 25
Crimping Tools	Pass-through RJ45 modular crimper, BNC compression/hex crimper, Bootlace ferrule crimping tool (four-mandrel design).	Sessions 1, 6, 13, 14
Conduit & Install Tools	20mm PVC pipe cold-bending spring, mini hacksaw, deburring file, 15m nylon draw-wire (fish tape), measuring tape.	Sessions 7, 15, 23
Heavy Power Tools	Impact hammer drill (reversible), 6mm concrete SDS-plus drill bits.	Sessions 15, 25
Basic Electrical Testers	Digital Multimeters (DMM with safety probes & leads), AC voltage test pen (non-contact), electrical series test lamp board.	Sessions 1, 2, 3, 4, 5, 24, 28
Network Testers	Digital wiremap cable tester (RJ45/RJ11 mapping), inline PoE voltage/wattage detector, managed switch status console.	Sessions 11, 13, 28
Fiber & CCTV Testers	Optical power meter (OPM), fiber cleaner click pen (1.25mm/2.5mm), handheld analog video signal level meter.	Sessions 12, 14, 28
Safety Equipment	LOTO lockout safety padlocks, hazard tags, insulated electrician safety gloves, protective eyewear.	Sessions 2, 4, 12

Master Bill of Materials & Consumables

Consolidated shopping list of all hardware modules, switches, cables, and consumables required to supply the lab workstations.

Category	Item Description	Primary Sessions Used
Electrical Modules	SPDT 6A switches, light holders, 60W incandescent lamps, Single-Phase DIN-rail SPD, SP MCB (6A), RCCB (30mA), 12V 10A multi-channel SMPS, 12V 7Ah SLA backup battery, 12V DC cooling fan, 12V relay modules.	Sessions 1, 2, 4, 5, 6, 21, 22
Lock Hardware	12V DC Electromagnetic Lock (maglock, 280kg holding force).	Session 6
Network Hardware	L2 Managed Network Switch, 8-Port 802.3at PoE switch (120W budget), SOHO Wi-Fi router, 10/100/1000 copper-to-fiber media converters.	Sessions 8, 9, 10, 11, 12, 19, 20, 26, 27
Fiber Components	Gigabit SFP fiber transceivers (Single-mode LC), LC-LC Duplex optical patch cords.	Session 12
CCTV Hardwares	Analog BNC test camera, IP dome camera (IK10), AI bullet IP camera, ANPR IP camera, ColorVu IP camera, Bi-spectrum thermal camera, Network PTZ dome camera, 4-Channel PoE NVR recorder, 4-Bay NVR or NAS server.	Sessions 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28
Storage Drives	Identical 2TB Surveillance-grade SATA HDDs (WD Purple or Seagate SkyHawk), hot-swap drive trays.	Sessions 17, 18
Passive Filters & Baluns	BNC passive UTP video baluns, video ground loop isolator filters.	Sessions 14, 28
Cables & Consumables	RG59 coaxial cable spool, Cat6 UTP bulk cable spool, RJ45 pass-through plugs, RJ11 modular plugs, connector rubber boots, bootlace wire ferrules, ferrule number tags, electrical insulation tape, twin-lead alarm wire, PVC solvent cement, concrete wall anchor plugs, mounting screws, cable zip-ties.	Sessions 1, 6, 7, 13, 14, 15, 21, 22, 28

Module 1: Electrical & Networking Basics

Session 01: Electrical Fundamentals (AC vs. DC)

Lab Goal: Wire a fully functional staircase (two-way) lighting loop.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Identify Phase, Neutral, and Ground terminals.	• 2x SPDT switches (6A)	• Wire stripper
2. Wire two SPDT (3-way) switches in a stairs toggle configuration.	• 1x light bulb holder	• Phillips & flat screwdriver
3. Connect travel lines between switches, route output to light holder.	• 1x 60W bulb	• Phase tester pen
4. Energize loop, verify dual-location light bulb toggling.	• Red, Black, Green wire (1.5mm ²)	• Electrical insulation tape
	• Wood/PVC wiring baseboard	

Session 02: Earthing & Surge Protection

Lab Goal: Verify earth continuity and measure voltage drops across Surge Protective Devices (SPDs).

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Perform visual inspection of grounding busbar connections.	• 1x Single-Phase SPD module	• Digital Multimeter (DMM)
2. Construct test lamp circuit; connect between Phase and Earth to check trip response.	• Copper earth clamp	• Earth ground loop resistance tester
3. Measure AC potential between Neutral-Earth (limit to <2V).	• 100W incandescent test bulb	• Insulated safety gloves
4. Install surge protector, verify grounding wire paths.	• Green ground cable (4mm ²)	

Session 03: Multimeter & Diagnostics

Lab Goal: Measure voltage, resistance, and troubleshoot short circuits.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Configure DMM selector dial to measure AC mains outlet voltage safely.	• 9V battery & 12V DC adapter	• 2x Digital Multimeters
2. Measure DC potentials on batteries and power adapters.	• Cartridge glass fuses (2A)	• Alligator clip leads
3. Use continuity mode to map a hidden cable bundle.	• Test resistors (1k Ω , 10k Ω)	• Wire cutters & strippers
4. Locate open and short circuit faults in a test wiring board.	• Hidden junction mock wire box	

Session 04: Electrical Safety & Breakers

Lab Goal: Test MCB overload tripping and RCCB earth leakage trip points.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Wire a mock distribution panel with a main MCB and RCCB. 2. Apply LOTO locks and safety tagout tags on the circuit breaker. 3. Introduce dynamic loads to verify MCB circuit protection. 4. Press RCCB test button, and measure leakage trip time manually.	• 1x SP MCB (6A) • 1x RCCB (30mA trip) • Lockout padlock & safety tag • Variable load board (bulbs)	• Insulated screwdrivers • DMM with current clamp • Series test lamp board

Session 05: Power Supply & SMPS

Lab Goal: Calibrate SMPS DC potentials and audit battery backup load runtimes.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Wire 230V AC mains into SMPS input terminals. 2. Adjust trimmer potentiometer to calibrate DC output to exactly 13.8V. 3. Connect 12V SLA backup battery to SMPS charging lines. 4. Cut AC power, monitor battery discharge curves under DC loads.	• 12V 10A multi-channel SMPS • 12V 7Ah SLA backup battery • 12V DC cooling fan (test load) • Red/Black DC cables	• Precision trimmer screwdriver • Multimeter • Crimping tool for spade lugs

Session 06: Reading Drawings (Relay & Lock Wiring)

Lab Goal: Wire an electromagnetic lock using relay NC/NO dry contacts.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Decode terminal legend symbols on access control drawings. 2. Verify relay dry contact vs wet contact jumper states. 3. Wire a 12V maglock to relay normally closed (NC) contacts. 4. Trigger relay coil to break loop, verifying lock release.	• 12V Electromagnetic Lock • 12V DC Relay module board • Bootlace ferrules & numbers • 12V DC power adapter	• Ferrule crimping tool • Precision screwdrivers • Wire stripper

Session 07: Cable Selection & Conduits

Lab Goal: Bends PVC conduit pathways and runs low-voltage lines.

Step-by-Step Lab Plan

1. Calculate wire fill ratios for 20mm PVC pipe paths.
2. Insert bending spring inside PVC pipe, execute a clean 90° cold bend.
3. Mount conduit saddles to wallboard, secure pipe runs.
4. Use draw wire to pull Cat6 UTP cables through bent pipe paths.

Materials Required

- 20mm PVC conduit pipes
- Conduit saddles & wall plugs
- Cat6 UTP cables
- PVC solvent cement

Tools to Teach

- 20mm Bending spring
- Hacksaw & deburring knife
- Nylon fish tape (draw wire)
- Measuring tape

Session 08: IP Addressing & Subnetting

Lab Goal: Map static subnets and assign manual IP grids to local clients.

Step-by-Step Lab Plan

1. Calculate IP address ranges for a CIDR /28 subnetwork.
2. Configure manual IP address and subnet mask on test computer.
3. Configure default gateway path to routing hub.
4. Run ping tests to verify local LAN interface connectivity.

Materials Required

- 2x Laptops / PCs
- Ethernet patch cables
- Subnetting calculation worksheets

Tools to Teach

- Windows CMD Terminal
- Wireshark packet analyzer

Session 09: Router & Switch Configuration

Lab Goal: Setup LAN subnets, configure WAN ports, and port forward services.

Step-by-Step Lab Plan

1. Log into network router administrative GUI panel.
2. Configure WAN port for static IP internet access.
3. Set up NAT port forwarding rules to link HTTP port 80.
4. Configure secure Wi-Fi SSID access profiles and WPA3.

Materials Required

- SOHO Wi-Fi Router / AP
- Ethernet patch cables
- Client testing laptop

Tools to Teach

- Web browser console
- Mobile Wi-Fi scanning app

Session 10: VLAN Basics

Lab Goal: Segment traffic using Layer 2 managed switch VLANs.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Connect client laptop to switch configuration console. 2. Define VLAN 10 (CCTV) and VLAN 20 (Office) on switch database. 3. Assign specific switch physical access ports to VLAN groups. 4. Configure 802.1Q tagged trunk links between switches.	• L2 Managed Network Switch • Ethernet console patch cords • 2x test client laptops	• PuTTY / Terminal Console • Switch CLI database

Session 11: PoE & PoE Switches

Lab Goal: Calculate PoE power budgets and deploy splitters/injectors.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Audit maximum switch wattage budget from manufacturer specifications. 2. Calculate aggregate camera draws including startup IR peaks. 3. Deploy passive PoE injector to power remote IP camera. 4. Use PoE splitter module to power a 12V non-PoE device.	• 802.3at PoE Switch • Passive PoE injector block • 12V DC active PoE splitter • Cat6 cables	• Inline PoE Wattage Tester • Managed switch web utility

Session 12: Fiber Optic Basics

Lab Goal: Inspect SFP fiber transceiver links and media converters.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Insert SFP transceivers into managed switch ports. 2. Inspect LC fiber connectors using standard optical cleaner. 3. Connect LC optical patch cord between SFP switch ports. 4. Deploy fiber-to-copper media converter, verify copper LAN bridge.	• 2x Gigabit SFP modules • LC-LC Duplex fiber patch cord • Fiber-to-Copper media converter • Fiber cleaning click pen	• Optical Power Meter (OPM) • Switch status indicators console

Session 13: Network Troubleshooting (RJ45 / RJ11 Crimping)

Lab Goal: Crimp Cat6 UTP cables and diagnostic wiremap failures.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Strip Cat6 outer sheath, separate twisted pairs onto spline. 2. Arrange wire strands in T568B sequence, crimp RJ45 plugs. 3. Strip phone cable, crimp RJ11 plug in 6P4C central configuration. 4. Insert both cables in wiremap tester, audit for split/reversed pairs.	• Cat6 UTP bulk cable spool • RJ45 pass-through plugs • RJ11 modular plugs • Connector rubber boots	• Pass-through RJ45 crimper • UTP cable sheath stripper • Digital wiremap cable tester

Module 2: Professional CCTV Systems

Session 14: Analog & IP CCTV

Lab Goal: Connect coaxial BNC feeds, video baluns, and compare network video lag.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Crimp 3-piece BNC connector on RG59 coax, link analog camera to XVR. 2. Install video baluns to run analog video over a Cat6 UTP cable. 3. Connect IP camera to switch, configure NVR display feed. 4. Display stopwatch on monitor, capture lag differentials.	• RG59 coax cable spool • 3-piece crimp BNC plug • Passive UTP video baluns • Analog & IP test cameras	• Coaxial cable rotary stripper • BNC hex crimping tool • Video signal level meter

Session 15: Camera Selection & Design

Lab Goal: Secure wall mount plates and adjust True WDR highlights.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Mount waterproof junction boxes to a masonry test wallboard. 2. Secure bullet and dome cameras onto corresponding bracket plates. 3. Aim camera at a high-contrast window backlight. 4. Navigate camera web console, toggle WDR to check face details.	• Dome & Bullet camera shells • Junction boxes & wall anchors • Masonry anchor plugs • Screws & gaskets	• Impact hammer drill • Concrete drill bits (6mm) • IP camera config tool

Session 16: Lens & Field of View

Lab Goal: Calibrate focal lengths and compare zoom resolution degradation.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Adjust zoom and focus rings on manual varifocal lens. 2. Use NVR zoom control panel to drive motorized varifocal cameras. 3. Deploy 2.8mm fixed camera, measure blindspots at entryways. 4. Magnify target details to compare optical vs digital resolution.	• Manual varifocal lens camera • Motorized zoom auto-focus IP camera • Testing target charts	• Camera focus adjustment tool • NVR PTZ control interface • Laser distance meter

Session 17: DVR, NVR & Hybrid Recorders

Lab Goal: Mount SATA storage disks and configure recording profiles.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Remove NVR cover, mount surveillance HDD using chassis screws. 2. Connect internal SATA data ribbon and motherboard power cords. 3. Boot recorder, format disk, verify active storage status. 4. Set up continuous vs. motion-triggered storage schedules.	• 4-channel PoE NVR unit • 2TB Surveillance HDD • SATA cable ribbons • Chassis screws	• Insulated Phillips screwdrivers • DMM (to check HDD power rails)

Session 18: RAID & Storage Design

Lab Goal: Configure RAID 5 array blocks and execute hot-swap disk rebuilds.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Access array management console under NVR configuration. 2. Initialize 3-disk RAID 5 storage array. 3. Simulate disk fault by physically unplugging a single HDD tray. 4. Hot-swap a clean replacement disk, monitor raid reconstruction.	• Multi-Bay NVR / NAS Server • 3x Identical 2TB SATA HDDs • Hot-swap drive tray caddies	• Server management web console • HDD tray release key

Session 19: ONVIF Standards

Lab Goal: Discover and integrate third-party IP cameras.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Install ONVIF Device Manager utility software on PC. 2. Scan subnet to discover third-party camera IP and connection port. 3. Enable ONVIF user profiles on camera firmware page. 4. Add camera to main NVR using the custom ONVIF profile, verify feeds.	• Third-party brand IP camera • Test client computer • Managed PoE network switch • NVR recorder	• ONVIF Device Manager software • IP scanner search tools

Session 20: Video Compression

Lab Goal: Measure and optimize H.264, H.265, and H.265+ data streams.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Configure camera compression format to H.264 profile. 2. Expose camera lens to high-motion environment, measure peak bitrates. 3. Change compression profile to H.265, check bandwidth savings. 4. Activate H.265+ smart codecs, monitor bitrates during static periods.	• High-resolution IP Camera (4K) • PoE switch & testing laptop • DC power adapter	• Router bandwidth monitor interface • Camera stream statistics utility

Session 21: AI Analytics & Perimeter Security

Lab Goal: Configure tripwires and wire hardware alarm relay outputs.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Navigate to smart event settings on AI camera web interface. 2. Draw a line-crossing boundary, configure directional filters. 3. Wire NVR alarm block outputs to a 12V warning strobe lamp. 4. Cross tripwire path, verify strobe lamp fires.	• AI classification IP camera • 12V DC Warning strobe / Siren • 12V DC Power supply adapter • Twin core alarm wires	• Precision terminal screwdrivers • Multimeter

Session 22: ANPR (Plate Recognition)

Lab Goal: Set fast exposure shutters and connect barrier gates.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Mount ANPR camera at lane entry template. 2. Adjust camera shutter speed manual profiles (1/1000s). 3. Import whitelisted test plates into camera database. 4. Wire camera control relay to barrier motor to verify gate trigger.	• ANPR optical IP camera • Printed mock license plates • 12V alarm indicator lamp • Twin-lead connection wire	• ANPR web configuration tool • Multimeter (continuity check) • Small flathead terminal screwdriver

Session 23: Facial Recognition

Lab Goal: Audit facial pixel density and build enrollment database blocks.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Mount face capture camera at gateway entrance area. 2. Use measuring tape to find eye pixel spacing on capture frames. 3. Enroll user profile images into Staff and Blacklist databases. 4. Test facial matching thresholds, verify blacklist alarm warnings.	• Face recognition IP camera • Enrollment photos (Staff, Blacklist) • Testing target cards	• NVR face matching console • Target pixel analysis calculator • Measuring tape

Session 24: ColorVu, AcuSense & Thermal

Lab Goal: Configure low-light warm LEDs and calibrate thermal heat sensors.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Configure supplemental warm light thresholds on ColorVu camera. 2. Adjust thermal sensor temp scale limits to detect hot spots. 3. Calibrate temperature alarm rules (e.g. trigger if over 60Â°C). 4. Simulate fire hazards using heat sources to verify alarms.	• ColorVu IP Camera • Bi-Spectrum thermal camera • Controlled heat source (soldering iron)	• Thermal imaging software • Non-contact laser thermometer

Session 25: PTZ Cameras

Lab Goal: Save PTZ coordinate presets and construct patrol sequence tours.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Control speed dome movement using NVR joystick controls. 2. Set up and save 4 separate preset coordinate locations. 3. Combine presets into a patrol tour sequence with 10s dwell times. 4. Enable smart auto-tracking features to trace target movement.	• Network PTZ dome camera • PTZ ceiling mount bracket • Twin core control wire (RS485)	• 3-Axis PTZ Joystick keyboard • NVR PTZ preset menu

Session 26: Remote Viewing & Cloud

Lab Goal: Set up P2P cloud accounts and optimize multi-stream bandwidth profiles.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Register NVR device serial number on cloud app server. 2. Scan screen QR code to bind recorder to a test mobile phone. 3. Expose camera, adjust Main stream and Sub stream bitrates. 4. Test WAN remote viewing loading times over cellular data networks.	• NVR recorder (connected to WAN) • Smartphone with remote app installed • Network patch cables	• Cloud P2P management portal • Bandwidth speed test utility

Session 27: Cybersecurity

Lab Goal: Secure NVR ports, install HTTPS certificates, and configure IP limits.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Disable UPnP auto-routing functions inside router. 2. Re-map standard connection ports (HTTP 80 and SDK 8000) to high ports. 3. Enable HTTPS protocol, load encryption certificates. 4. Configure IP whitelisting parameters for admin remote access.	• Network NVR system • Testing laptop • Network switch	• NVR network hardening settings menu • SSL certificate creation console

Session 28: Troubleshooting & Maintenance

Lab Goal: Fix video loss errors, resolve IP clashes, and clean dome cameras.

Step-by-Step Lab Plan	Materials Required	Tools to Teach
1. Isolate duplicate IP address conflict issues using ping scans. 2. Install ground loop isolator to block analog video hum bars. 3. Inspect and clean dirty camera glass to prevent night IR halos. 4. Recover locked firmware using TFTP automated network boot tools.	• Ground loop isolator filters • Camera lens cleaning kit • NVR/Camera firmware update bin files	• TFTP desktop server utility • IP scanning software tool • Lens blower & clean microfiber cloth